

EEBus SHIP Implementation Guideline

Transport and Connectivity

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1 Scope of the document

The Smart Home IP (SHIP) specification defines the secure transport layer for the EEBUS communication stack. It utilizes established IT standards such as mDNS, TLS and WebSockets to ensure secure, IP-based point-to-point communication between devices.

While the SHIP specification provides the formal requirements for connection establishment and data framing, real-world implementations often face challenges regarding network topologies, certificate handling and edge cases during the handshake process.

The purpose of this Implementation Guideline (IG) is to provide developers and integrators with supplementary clarifications, best practices and strict interpretations of the SHIP specification. By following this guideline, implementers can avoid common pitfalls, ensure a robust connection lifecycle and guarantee interoperability during the certification process.

1.1 References

1.1.1 EEBUS documents

[SPINE] *Minimum:* EEBus_SPINE_V1.3.0_Final.zip

[SHIP] *Minimum:* EEBus_SHIP_TS_Specification_v1.0.1.pdf

Recommended: EEBus_SHIP_TS_Specification_v1.1.0.pdf

1.1.2 Normative references

[RFC2119] IETF RFC 2119: 1997, Key words for use in RFCs to indicate requirement levels. Please see section 1.3.1 for details.

[RFC8259] <https://datatracker.ietf.org/doc/html/rfc8259>

1.2 Terms and definitions

IG-SHIP

Abbreviation for "Implementation Guideline for SHIP" (short name of this document)

SME

Abbreviation for "SHIP Message Exchange"

SPINE

Abbreviation for "Smart Premises Interoperable Neutral-message Exchange": Technical Specification of EEBus Initiative e.V.

1.3 Requirements

1.3.1 Requirements wording

The following keywords are used:

- SHALL
- SHALL NOT
- SHOULD
- SHOULD NOT
- MAY

Note: They apply only if written in capital letters.

For the meaning of the keywords, please refer to [RFC2119].

1.3.2 Notes on requirements in this document

This document clarifies requirements of the [SHIP] Specification.

Regarding these specifications, the keywords "SHALL" and "SHALL NOT" are generally already covered by mandatory requirements of these specifications. However, in cases where this guideline explicitly defines retroactive changes to the existing specification, these keywords express mandatory requirements that intentionally override the original specifications. Such modifications are additionally tracked via errata sheets.

2 Clarifications on the SHIP Specification

2.1 Parallel processing of access methods and SPINE messages

After the connection establishment process (including the TLS handshake and initial SME phases), the connection enters the "connection data exchange" state. In this state, devices can exchange both SHIP-specific messages (such as an `accessMethodsRequest`) and application-layer SPINE messages concurrently.

While querying the `accessMethods` is an optional procedure (though answering a received request is mandatory), it frequently occurs simultaneously with the initial SPINE discovery phase (e.g., the exchange of `nodeManagementDetailedDiscoveryData` or `nodeManagementUseCaseData`).

Avoiding message exchange blockades:

A wrong behaviour would be if Device A sends an `accessMethodsRequest` and subsequently blocks or ignores all incoming SPINE messages until it receives the corresponding `accessMethodsResponse` from Device B. Such a behaviour could cause that Device B might run into timeouts for the SPINE responses it expects from Device A. This is especially problematic at the beginning of a connection where initial SPINE discovery messages might be required by Device B. Typically, these initial discovery messages are not re-transmitted, resulting in the assessment that the communication with Device A is currently not working properly. However, a restart of the communication could lead to the same situation.

Clarification and requirement:

To ensure robust interoperability and prevent initialization deadlocks, implementers SHALL adhere strictly to the following rules:

- No blocking: A device SHALL NOT block, drop or delay the processing of incoming connection data exchange messages (with e.g., SPINE payload messages) while waiting for an `accessMethodsResponse` (or any other optional SME response).
- Immediate readiness: As soon as the "connection data exchange" state is reached, the protocol stack SHALL be fully prepared to receive, parse and process SPINE messages independently of the current state of SME requests.
- Independent layers: When in state "connection data exchange", any incoming SPINE message SHALL be passed to the application layer immediately, even if the SME layer is still awaiting the resolution of an access methods query.
- Decoupled SME responses: A device receiving an `accessMethodsRequest` SHALL NOT delay or block its `accessMethodsResponse` while waiting for the remote device to process, acknowledge or reply to concurrent SPINE messages. The SHIP Message Exchange (SME) SHALL operate and respond independently of the application layer's progress.
- Defensive interoperability (legacy handling): To accommodate legacy devices in the field that might erroneously suspend their SPINE message processing until the SME access methods exchange concludes, devices SHOULD process and dispatch the `accessMethodsResponse` as

promptly as possible. This defensive approach ensures that constrained or strictly sequential remote peers are unblocked quickly, allowing the SPINE discovery phase to proceed.

2.2 Handling of whitespace in JSON payloads

While SHIP messages are typically transmitted in a compact format without unnecessary characters to minimize bandwidth, the underlying JSON standard allows for the inclusion of specific whitespace characters to improve human readability.

As defined in [RFC8259], whitespace is considered "insignificant" when it appears before or after any structural characters (such as {, }, [,], :, and ,).

Requirement for parsing:

To ensure robust interoperability, all EEBUS devices SHALL be capable of correctly interpreting and parsing SHIP JSON messages that contain any of the following whitespace characters:

- Space: (0x20)
- Horizontal tab: (0x09)
- Line feed (LF): (0x0A)
- Carriage return (CR): (0x0D)

Implementation note:

Devices SHALL NOT rely on the assumption that incoming JSON messages are delivered in a "minified" or "compact" format. Any JSON parser used within the SHIP stack SHALL be compliant with the whitespace rules of the JSON specification. Failing to handle these characters correctly is a violation of the specification and will lead to communication failures with devices that use formatted/pretty-printed JSON output.

2.3 On the usage of "auto accept"

As the "auto accept" mode for pairing SHIP nodes suffers from various security weaknesses, devices SHALL NOT use "auto accept" at any time.

This means that

- the device itself SHALL NOT activate "auto accept" mode on its own,
- the device SHALL NOT enable a user to activate "auto accept" mode,
- the "register" key in the SHIP TXT record SHALL NOT be set to "true" at any time.